

1 Introduction

Generative artificial intelligence (genAI) has quickly emerged as a potentially important workplace technology. The large language model ChatGPT debuted in November 2022 and, by December 2024, had over 300 million weekly users. Several studies find that genAI boosts productivity in specific cases (Brynjolfsson et al., 2023; Cui et al., 2024; Dell’Acqua et al., 2023; Demirci et al., 2025; Dillon et al., 2025; Kanazawa et al., 2025; Noy and Zhang, 2023; Otis et al., 2024; Peng et al., 2023), and some project genAI will have large macroeconomic effects (Chui et al., 2023; Filippucci et al., 2024). Yet others predict only modest aggregate impacts (Acemoglu, 2024). These differing views largely reflect uncertainty about the extent of genAI adoption today and in the future.¹

This paper studies the adoption of genAI using data from the first nationally representative U.S. surveys of genAI usage at work and at home. Our data come from the Real-Time Population Survey (RPS), a national online labor market survey of working age adults aged 18-64 that has run since 2020 (Bick and Blandin, 2023). We report the combined results of two surveys fielded in August and November 2024, which collectively included more than 10,000 respondents. These surveys include questions about genAI adoption that we adapt from the Current Population Survey’s Computer and Internet Supplement (CIU), which was introduced in 1984 to document computer (and later, internet) adoption. We also introduce novel questions to measure how intensively individuals use the technology, which tasks genAI assists with, how much time it saves them, and workplace barriers or support for adoption. The resulting microdata will be made publicly available, addressing calls for more research that develops and shares novel data (Ethiraj et al., 2019).

We use this data to address three main questions. First, how many individuals are currently using genAI? Pooling our August and November 2024 surveys, 32% of workers reported using genAI for work: 10% used it every work day, 17% on some but not all work days, and 5% did not use it in the previous week. One third of respondents report using genAI outside of work. Overall, 45% of respondents report using genAI either for work or outside of work. The most commonly used products are ChatGPT (32% of all respondents), Gemini (19%) and embedded products such as Microsoft Copilot (15%).

To benchmark these results, we show that the rate of workplace genAI adoption in 2024, two years after the release of ChatGPT, is similar to the rate of workplace computer adoption in

¹Acemoglu et al. (2022) analyze online job vacancies related to AI, but their data predates ChatGPT. Albanesi et al. (2024), Eisfeldt et al. (2024), and Gathmann et al. (2024) compare occupations according to predictions of AI exposure but do not have data on actual adoption. Humlum and Vestergaard (2025) find no meaningful impacts of genAI adoption on employment and earnings in Denmark among workers in 11 occupations that are particularly exposed to genAI.

rapidly (Nelson and Phelps, 1966). Third, genAI may directly increase workers’ productivity and wages. Our data do not allow us to distinguish between these explanations.

4.1 Tasks, Predicted Exposure, and Generative AI Adoption

Why do some groups of workers adopt genAI at higher rates than others? One reason could be that workers differ in their job tasks or responsibilities, and genAI is more helpful in assisting some tasks than others.

Consistent with this explanation, we first document that genAI adoption differs greatly between occupations; see Figure 5. The three occupations with the highest genAI adoption are computer and mathematical, management, and business and finance occupations (54%, 52%, and 48%, respectively) and the bottom three are blue collar, office and administration, and personal services occupations (26%, 23%, and 16%, respectively). At the same time, genAI adoption exceeds 20% in all but one occupation groups, suggesting that the technology is useful for a broad range of occupations. As with demographic differences, differences between occupations are driven by the rates of daily and weekly genAI users.

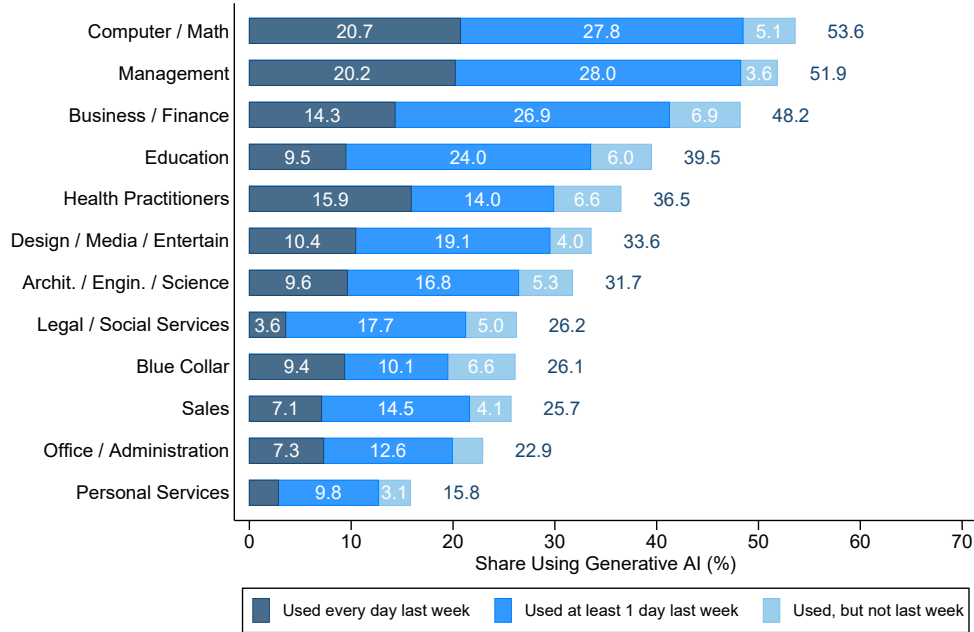
Do different adoption rates between occupations reflect differences in tasks or responsibilities? One approach to assess this is to construct task-based-exposure measures across occupations (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020). For example, Eloundou et al. (2024) constructs a rubric that assigns each task in the O*NET database a zero or one corresponding to whether LLMs (a popular class of genAI) could feasibly produce large productivity gains for that task. They then construct a predicted exposure score for each occupation by aggregating task-level scores for each occupation. Using RPS data, we can ask whether these predictions are correlated with adoption. Our study is the first to evaluate these task-based “exposure” predictions using actual genAI adoption rates.¹²

Figure 6a plots an occupation group’s predicted genAI exposure from Eloundou et al. (2024) against actual genAI adoption in the 2024 RPS. We find that predicted genAI exposure is highly correlated with actual genAI adoption ($\rho = 0.65$). In Figure 6b, we find a similar correlation ($\rho = 0.71$) using the predicted exposure score from Felten et al. (2021). For example, personal service and blue-collar occupations have relatively low predicted exposure and also relatively low adoption, while computer and mathematical occupations have both high predicted exposure and high adoption.

The high correlation between predicted exposure and actual adoption provides partial validation of task-based exposure predictions, supporting their utility for researchers. At the same

¹²Handa et al. (2025) assign tasks to conversations by Claude.ai Free and Pro users and show that task usage rates are correlated with the predictions by Eloundou et al. (2024).

Figure 5: Generative AI Work Adoption By Occupation



Notes: The figure shows adoption rates at work for genAI by occupation. The data source is the August and November 2024 waves of the RPS (blue bars). The sample is employed individuals age 18-64 (RPS, $N = 6883$). We exclude workers in “agriculture / forestry / fishing” due to a very low sample size. For details on occupation categories, see Appendix A.4.

time, we also note particular occupations whose adoption rates deviate from what we would expect based on their exposure score. In particular, office and administrative occupations and managers are predicted to be similarly exposed to genAI, but actual adoption by managers is more than double that of office and administrative workers. One potential reason for these discrepancies could be that existing task-based predictions do not perfectly identify which occupations would benefit from genAI.

To investigate this possibility, we leverage additional information about workers’ adoption decisions from the RPS. Among users who did not adopt genAI, we provided a list of potential reasons why they did not adopt. We then asked respondents to select the reasons that were relevant for their decision and to rank them in order of importance. We define the share of an occupation who does not use genAI because it is not useful for their job as the share of all workers in that occupation who ranked “GenAI cannot help me with my job” in their top two reasons for not using genAI. (For the full list of potential reasons, see Appendix A.5.)

Figure 6c plots genAI adoption against the share who reported that it was not useful. We find that occupational adoption is highly correlated (-0.88) with the share who reported genAI was not useful. For example, only 7% of workers in computer / math occupations reported that genAI was not useful for their job, compared with 30% of personal service workers.

6 Conclusion

Predictions about genAI’s economic impact vary widely, driven by uncertainty over the share of work it can assist and the size of its productivity potential. This paper presents evidence from the first nationally representative U.S. surveys on generative AI use at work and home. We find that a substantial share of individuals already use genAI. Thus far, the pace of overall adoption is rapid compared with past ICT technologies. At work, genAI adoption has followed a similar pace to PCs and is far ahead of genAI adoption by firms. We document wide variation in genAI use across demographic groups and occupations, and provide evidence that both task composition and firm policies appear to play an important role in workers’ adoption choices. Although the intensity of genAI use varies widely, genAI already assists a noticeable share of total work hours, suggesting the potential for meaningful productivity gains.

Our findings help lay the groundwork for understanding the labor market implications of genAI. They can also inform responses by managers and policymakers along several dimensions, such as data center and electricity needs, workforce training, and competitive dynamics.

References

- Abbate, J. (1999). “Getting small: a short history of the personal computer”. In: *Proceedings of the IEEE* 87.9, pp. 1695–1698.
- Acemoglu, D. (2024). *The Simple Macroeconomics of AI*. Working Paper. National Bureau of Economic Research.
- Acemoglu, D., D. Autor, J. Hazell, and P. Restrepo (2022). “Artificial intelligence and jobs: Evidence from online vacancies”. In: *Journal of Labor Economics* 40.S1, S293–S340.
- Albanesi, S., A. Dias da Silva, J. F. Jimeno, A. Lamo, and A. Wabitsch (Nov. 2024). *New Technologies and Jobs in Europe*. Working Paper. OIGI.
- Babina, T., A. Fedyk, A. He, and J. Hodson (2024). “Artificial intelligence, firm growth, and product innovation”. In: *Journal of Financial Economics* 151, p. 103745.
- Barrero, J. M., N. Bloom, and S. J. Davis (2021). *Why Working From Home Will Stick*. Working Paper. National Bureau of Economic Research.
- Bick, A. and A. Blandin (2023). “Employer reallocation during the COVID-19 pandemic: Validation and application of a do-it-yourself CPS”. In: *Review of Economic Dynamics* 49, pp. 58–76.
- Bick, A., A. Blandin, A. Caplan, and T. Caplan (2025a). “Heterogeneity in Work From Home: Evidence from Six U.S. Datasets”. In: *Federal Reserve Bank of St. Louis Review* 107.14, pp. 1–23.
- Bick, A., A. Blandin, A. Caplan, and T. Caplan (2025b). “Measuring Trends in Work From Home: Evidence from Six U.S. Datasets”. In: *Federal Reserve Bank of St. Louis Review* 107.15, pp. 1–23.
- Bonney, K., C. Breaux, C. Buffington, E. Dinlersoz, L. S. Foster, N. Goldschlag, J. C. Haltiwanger, Z. Kroff, and K. Savage (2024). *Tracking firm use of AI in real time: A snapshot*